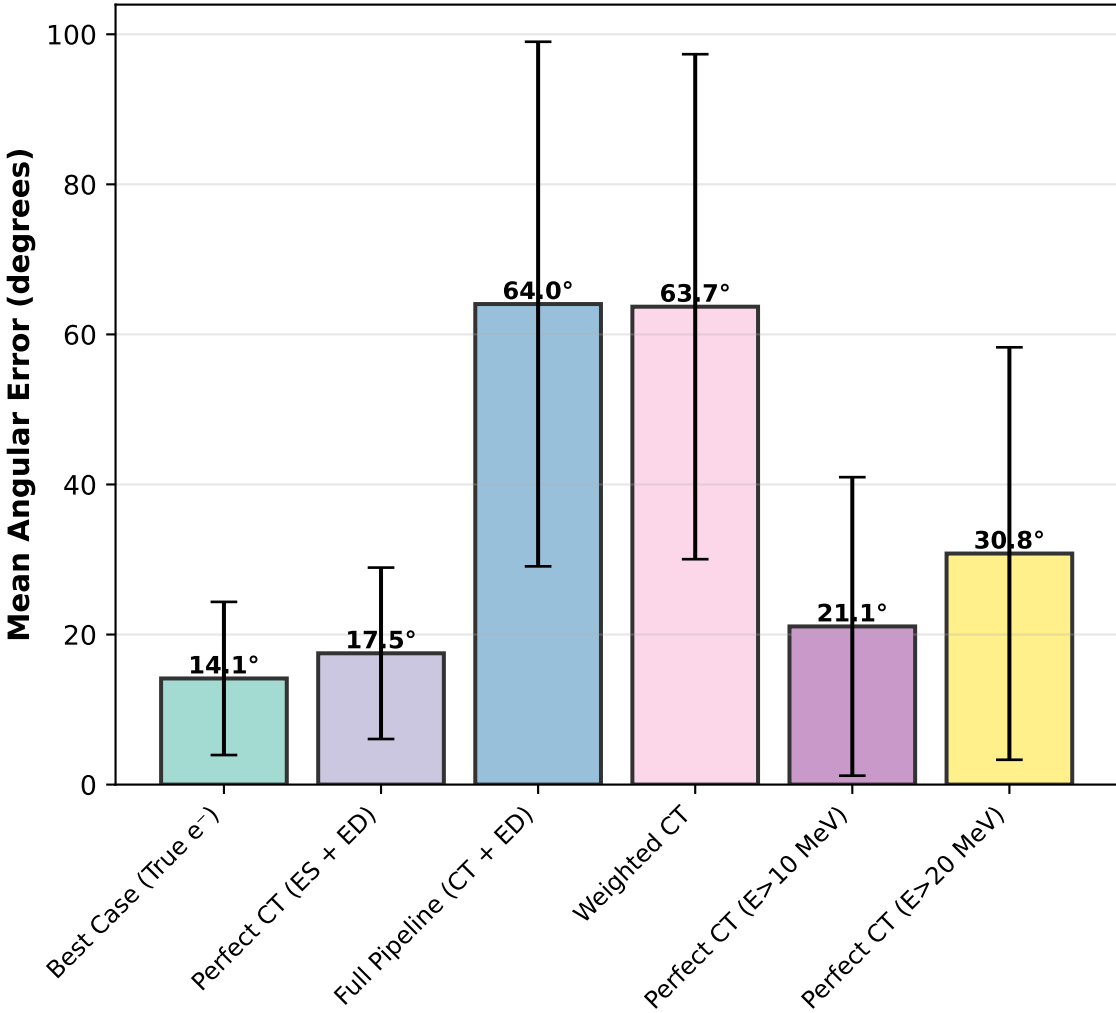


Emcee 6-Scenario Aggregate Analysis

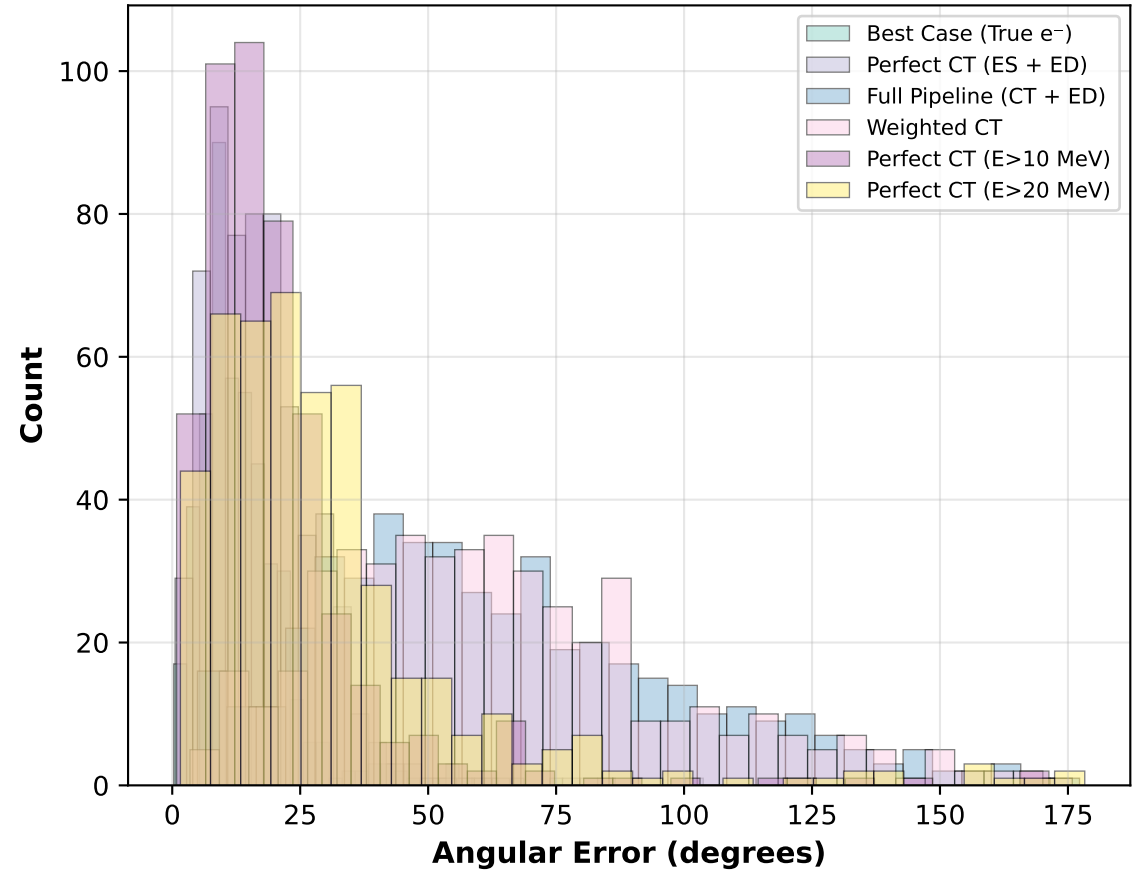
Summary Statistics

Scenario	N	Mean	Median	Std
Best Case (True e ⁻)	464	14.15°	11.85°	10.20°
Perfect CT (ES + ED)	615	17.50°	16.01°	11.42°
Full Pipeline (CT + ED)	464	64.05°	57.76°	34.96°
Weighted CT	464	63.69°	58.83°	33.65°
Perfect CT (E>10 MeV)	464	21.08°	16.76°	19.89°
Perfect CT (E>20 MeV)	464	30.79°	24.42°	27.49°

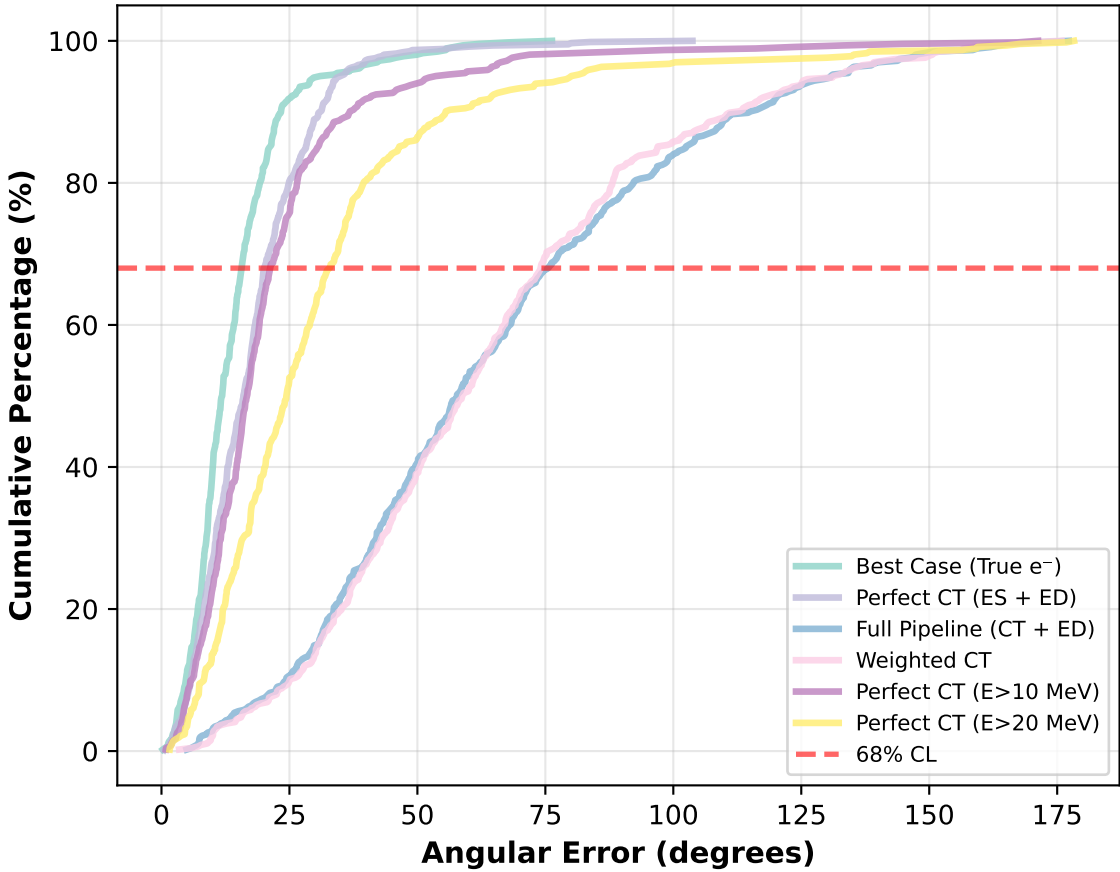
Mean Resolution ± Std Dev



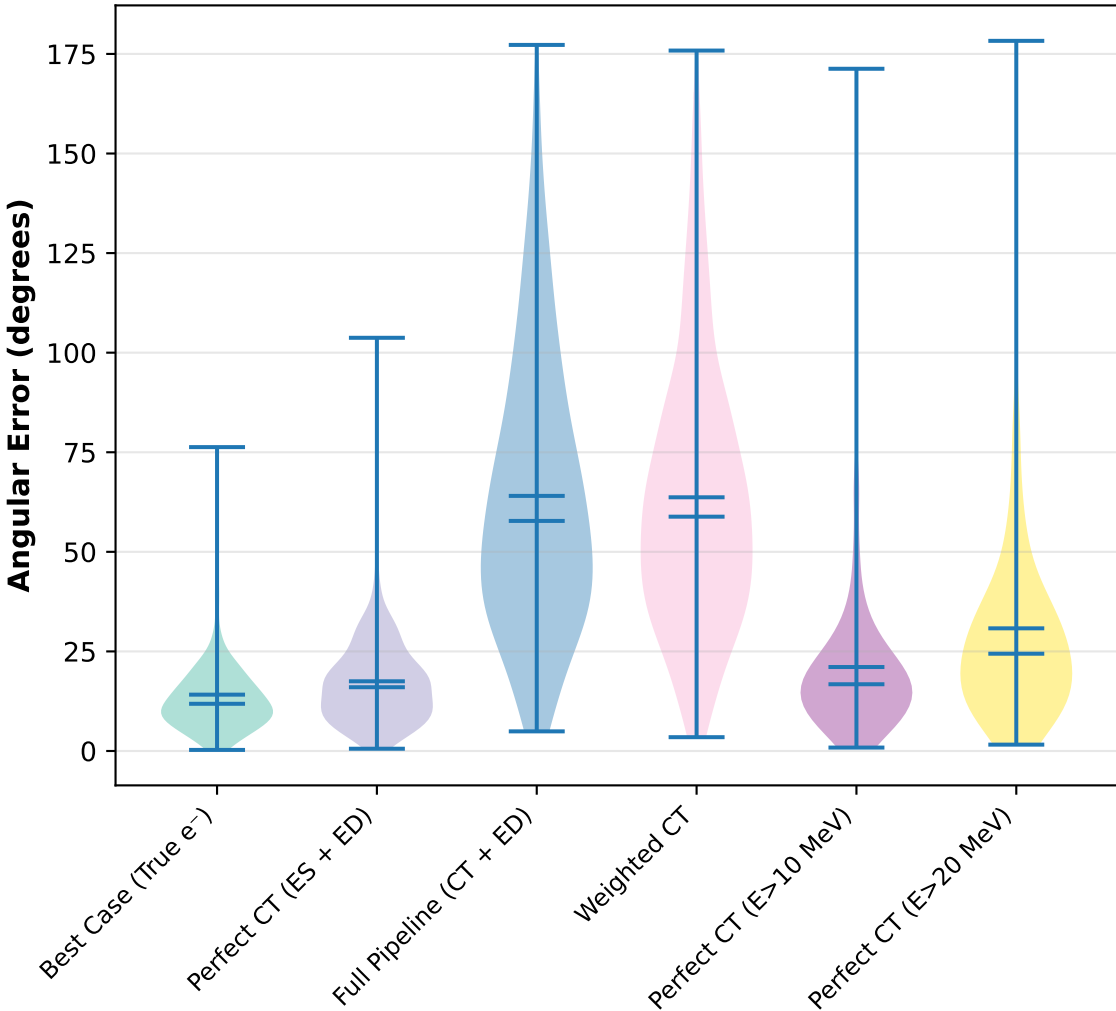
Distribution Overlay



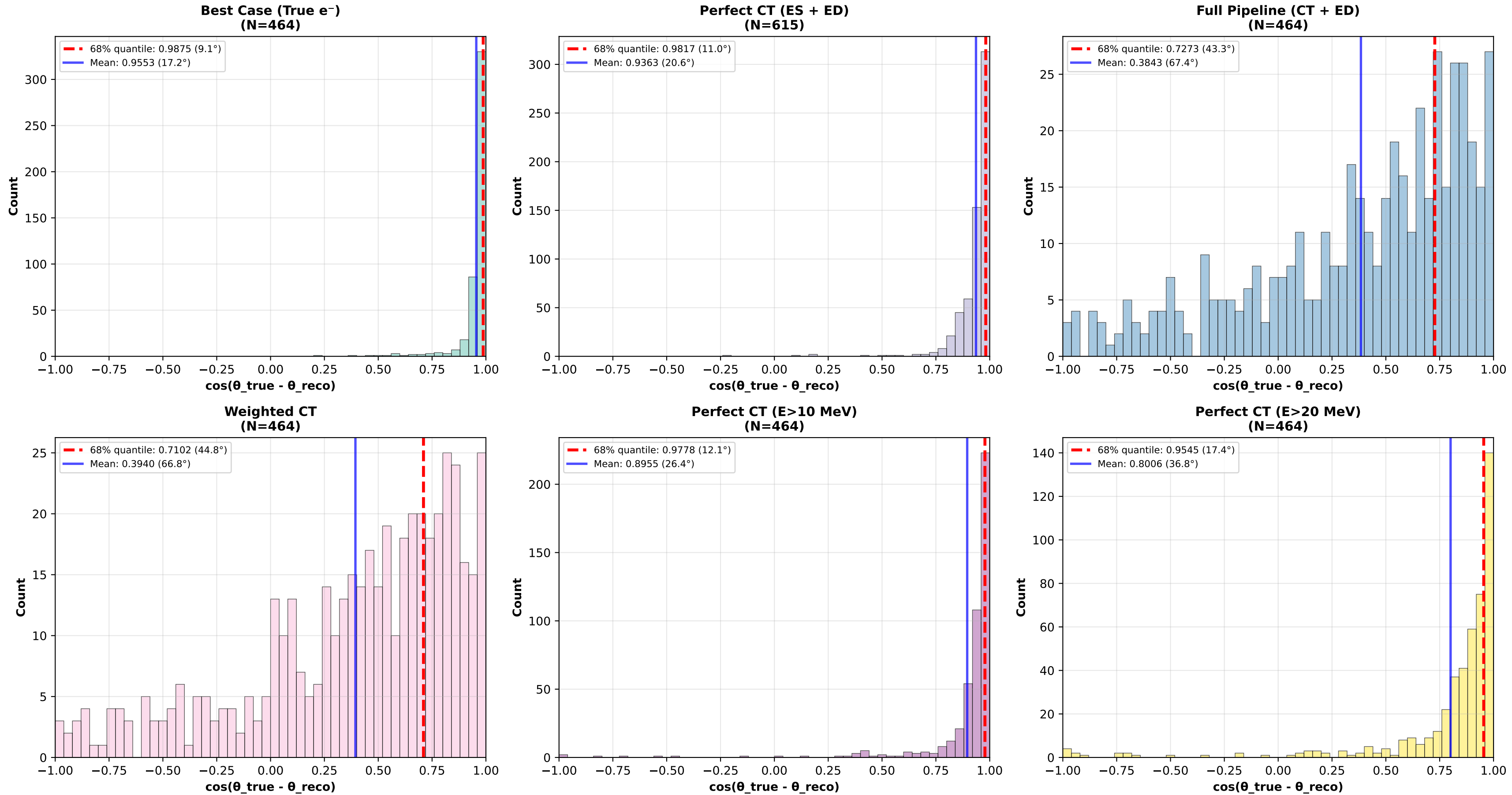
Cumulative Distribution



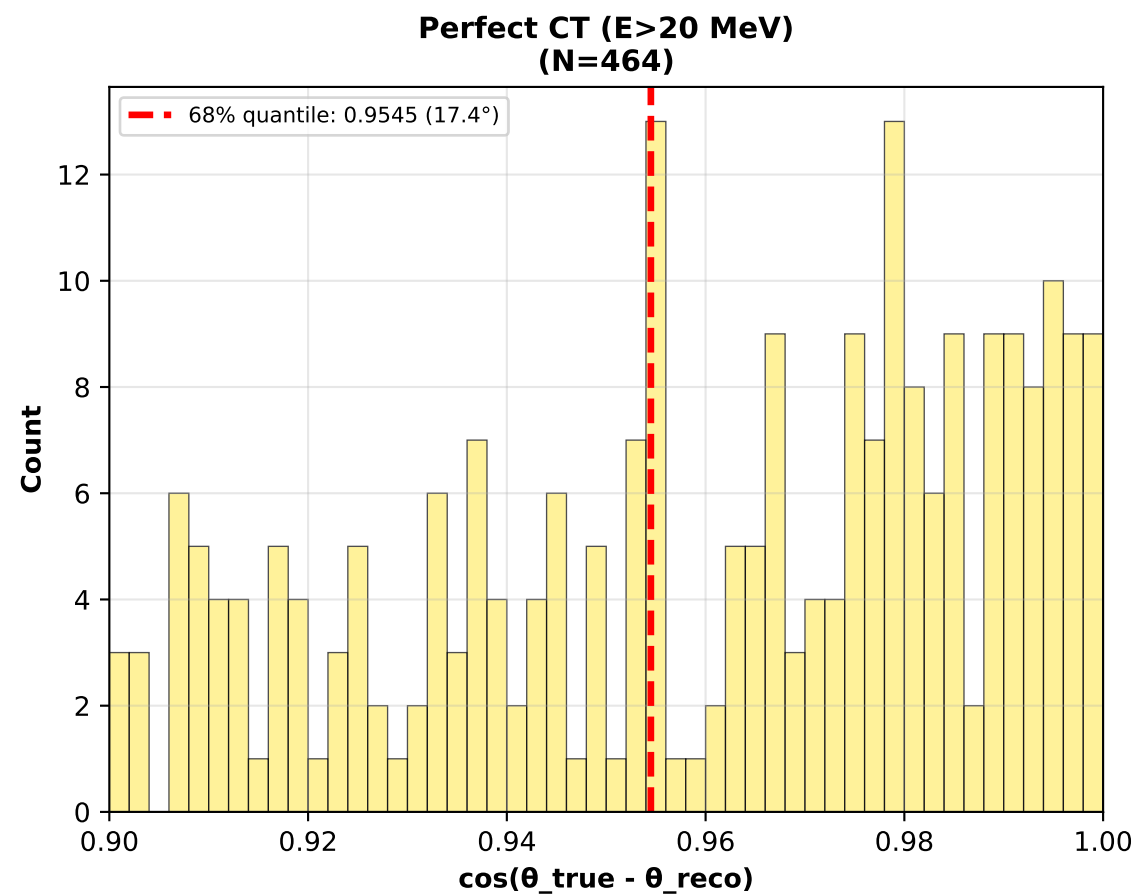
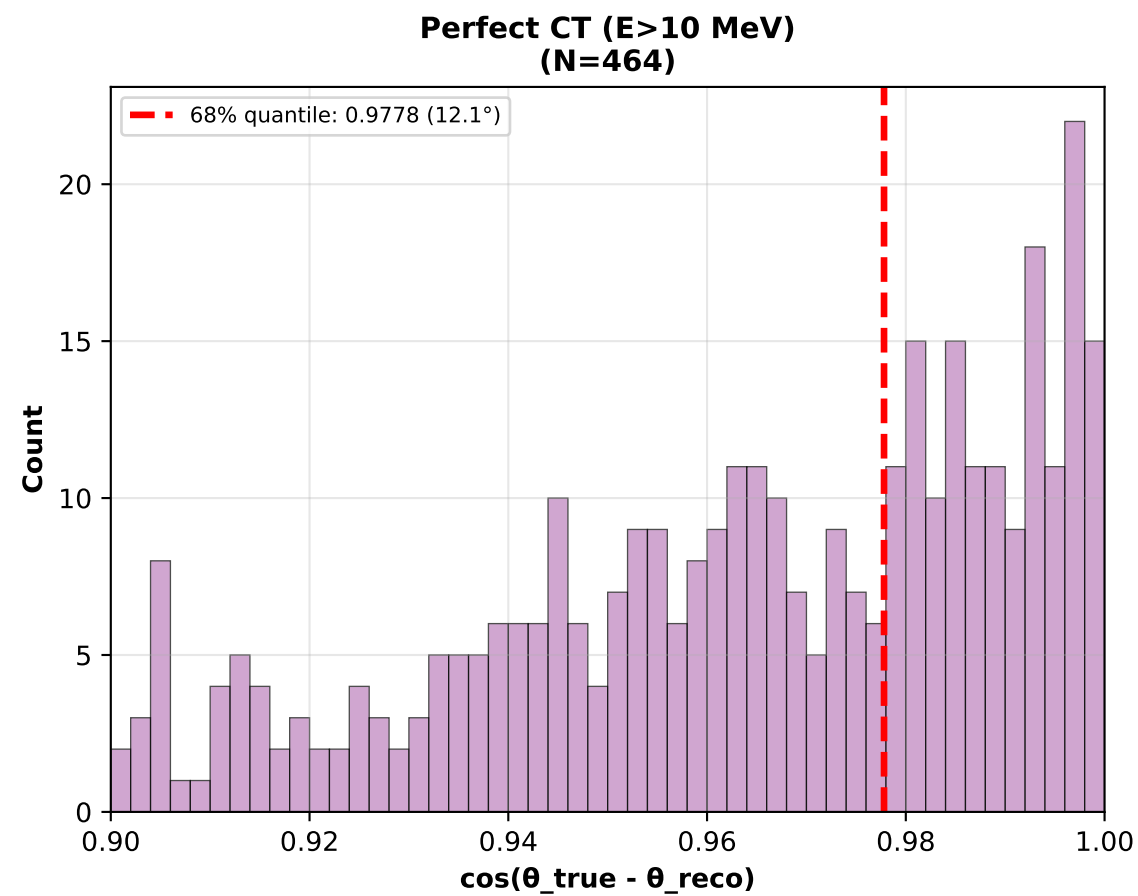
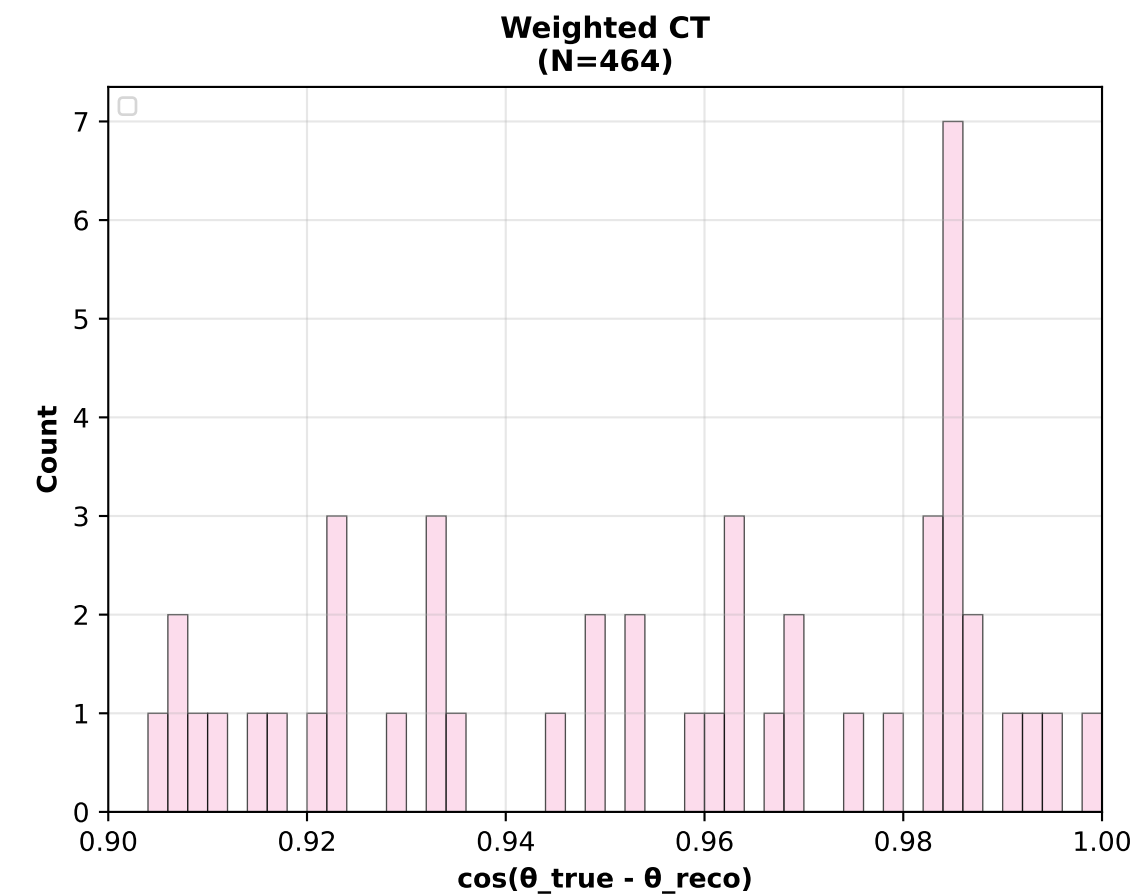
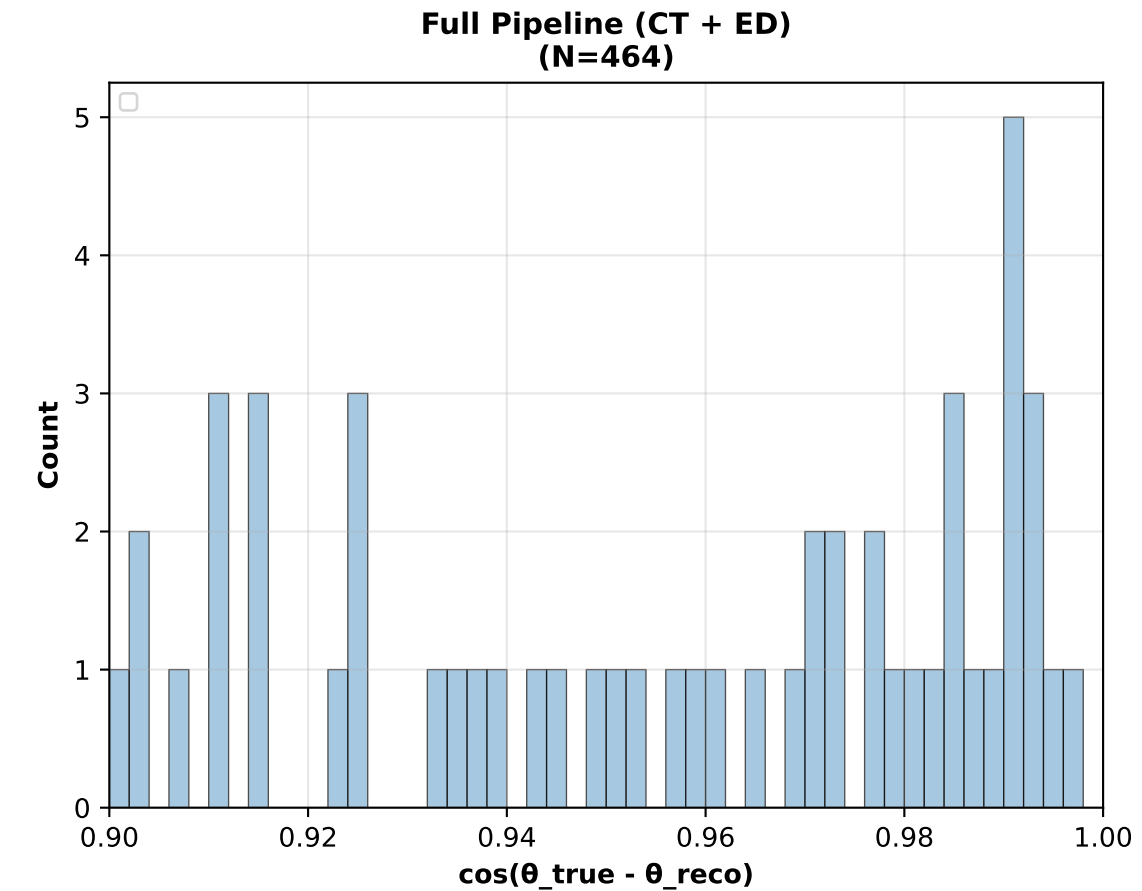
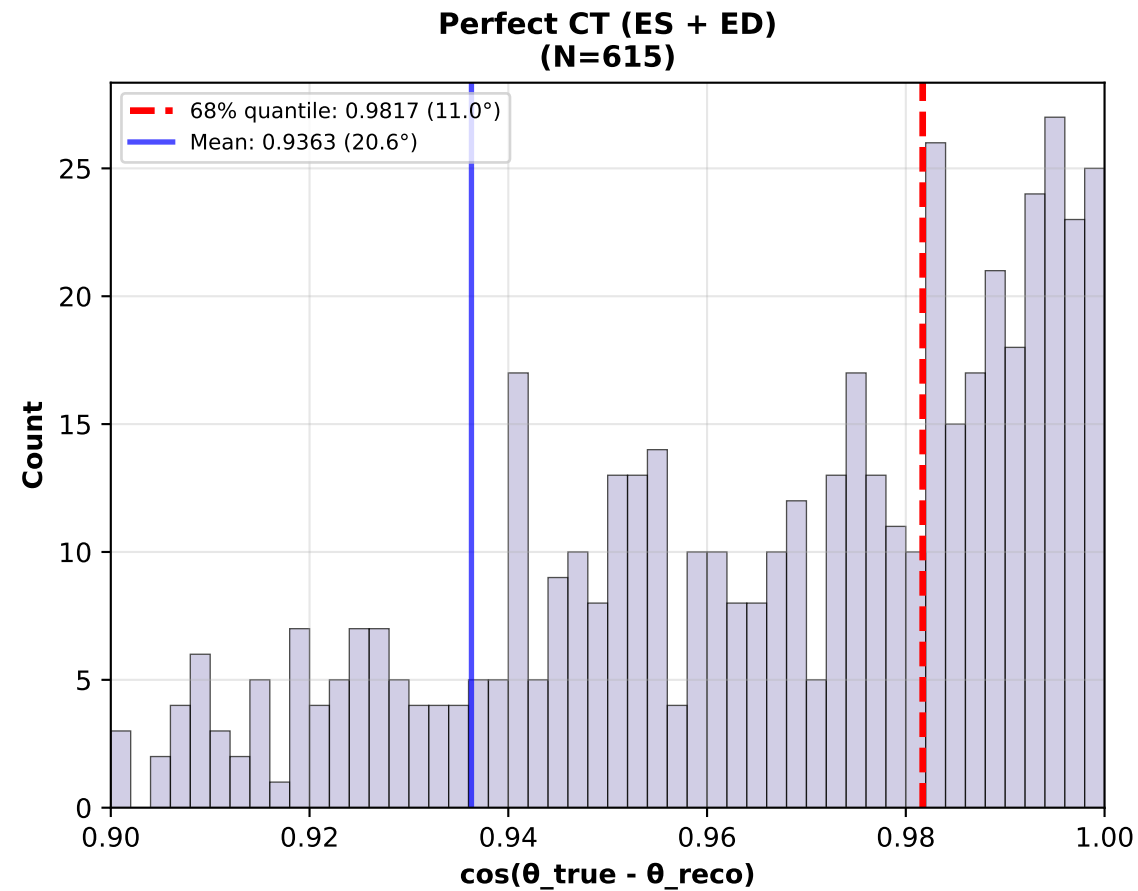
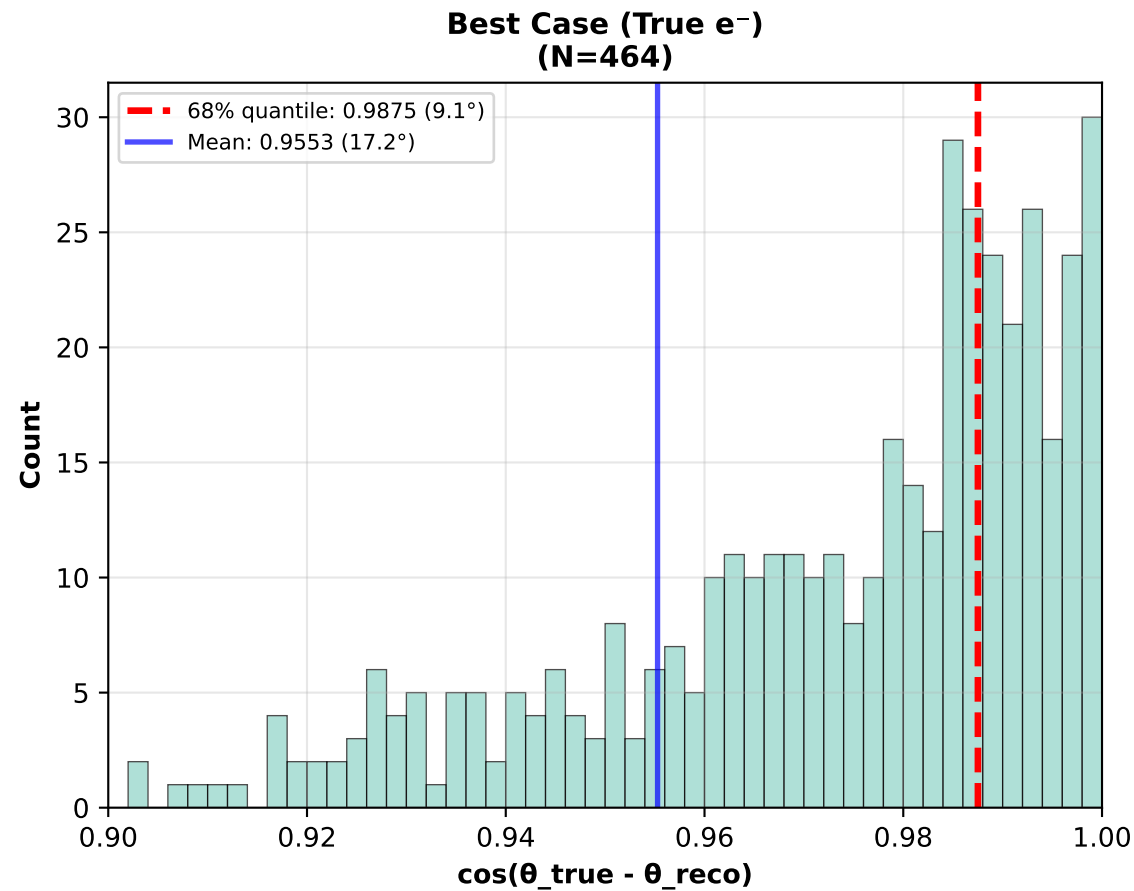
Distribution Shapes



Cosine Distribution: $\cos(\theta_{\text{true}} - \theta_{\text{reco}})$ - Full Range



Cosine Distribution: $\cos(\theta_{\text{true}} - \theta_{\text{reco}})$ - Zoomed (0.9-1.0)



ES Main Track Retention Across Scenarios

Scenario	N Cats	Input Total	Input ES Main	Clusters Used	Total Retention	ES Main Loss (Est.)
Best Case (True e ⁻)	464	308	29	266	86.3%	4
Perfect CT (ES + ED)	615	305	29	265	86.9%	4
Full Pipeline (CT + ED)	464	308	29	3245	1055.1%	-276
Weighted CT	464	308	29	3245	1055.1%	-276
Perfect CT (E>10 MeV)	464	308	29	144	46.7%	15
Perfect CT (E>20 MeV)	464	308	29	33	10.7%	26

*Note: "ES Main Loss (Est.)" shows estimated ES main tracks lost due to filtering/energy cuts.
For Full Pipeline and Weighted CT, losses are primarily from CT network misclassification.
For energy cut scenarios (E>10/20 MeV), losses are from energy threshold filtering.*

Cluster Usage Metrics

Scenario	N	Total In	ES In	Value Used	Note
Best Case (True e ⁻)	50	319	30	272	85% ret
Perfect CT (ES + ED)	50	319	30	272	85% ret
Full Pipeline (CT + ED)	50	319	30	3364	MCMC weighted
Weighted CT	50	319	30	3364	MCMC weighted
Perfect CT (E>10 MeV)	50	319	30	147	46% ret
Perfect CT (E>20 MeV)	50	319	30	34	11% ret